Matgeo-2.9.5

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Question

$|\mathbf{a}| = 3, |\mathbf{b}| = 2\sqrt{3},$ and $\mathbf{a} \cdot \mathbf{b} = 6$, then find the value of $|\mathbf{a} \times \mathbf{b}|$.

Solution

Given:

$$\|\mathbf{a}\| = 3 \tag{1}$$

$$\|\mathbf{b}\| = 2\sqrt{3} \tag{2}$$

$$\mathbf{a}^\top \mathbf{b} = 6 \tag{3}$$

We know:

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin\theta \tag{4}$$

$$\cos\theta = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \tag{5}$$

Solution

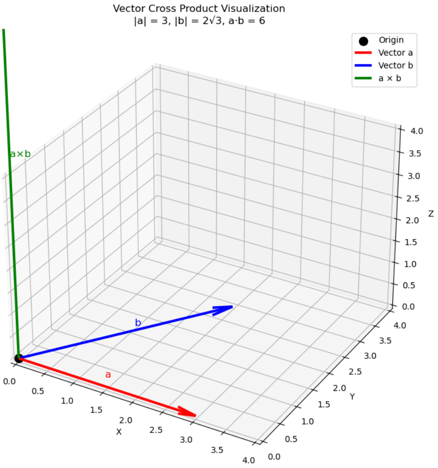
Thus

$$\left(\mathbf{a}^\top \mathbf{b}\right)^2 + (\|\mathbf{a} \times \mathbf{b}\|)^2 = \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 \tag{6}$$

Substituting values

$$\|\mathbf{a} \times \mathbf{b}\| = \sqrt{(3^2) \times (2\sqrt{3})^2 - 6^2} \tag{7}$$
$$\|\mathbf{a} \times \mathbf{b}\| = \sqrt{108 - 36} \tag{8}$$
$$\|\mathbf{a} \times \mathbf{b}\| = \sqrt{72} \tag{9}$$
$$\|\mathbf{a} \times \mathbf{b}\| = 6\sqrt{2} \tag{10}$$
$$\tag{11}$$

Plot



Vector Cross Product Visualization
|a| = 3, |b| = 2√3, a·b = 6
Origin
Vector a
Vector b
a × b
a×b
b
a
X
Y
Z
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

Figure